PETER CHIKA OZO-OGUEJI

peterchika3254@gmail.com
LinkedIn GitHub Portfolio
ORCID | Washington, DC | +1 (202) 977-1952

Data Scientist and Machine Learning Researcher
Machine Learning for Healthcare

RESEARCH INTERESTS

Trustworthy and continual machine learning for healthcare; continual medical anomaly detection; multimodal clinical AI; medical imaging; calibration, interpretability, and robustness under distribution shift.

EDUCATION

American University, Washington, DC *Aug 2023 - Dec 2025*
M.S. Data Science, GPA 3.83/4.00
M.S. Business Analytics and Artificial Intelligence, GPA 3.36/4.00
Selected coursework: Big Data Computing and Machine Learning, Neural Networks and Deep Learning, Natural Language Processing, Advanced Machine Learning, Statistical Programming in R, and Regression

Philippines College of Health and Sciences, Manila, Philippines *Aug 2019 - Aug 2021*
M.S. Mathematics

Federal University of Technology, Owerri, Nigeria *Sept 2010 - Dec 2015*
B.S. Statistics

RESEARCH EXPERIENCE

Continual Medical Image Learning, American University *Oct 2026 - Present*
Faculty mentor: Prof. Roberto Corizzo | Independent benchmark research
Developed a supervised, six-domain normal-versus-abnormal continual-learning benchmark using BMAD and Mammoth, with group-aware train/validation/test splits and leakage audits.
Compared fine-tuning, Experience Replay (ER), DER++, A-GEM, online EWC and Synaptic Intelligence (SI), with joint-training and per-domain references; evaluated three seeds, task order, AUROC and forgetting.
DER++ achieved final mean AUROC 0.915 ± 0.034 versus 0.612 ± 0.033 for fine-tuning across six tasks. Next direction: normal-only continual anomaly detection with PyCLAD; this extension is proposed, not a completed experiment.

Independent Study in Medical Machine Learning, American University *Aug 2025 - Dec 2025*
Advisor: Prof. Ahmad Mousavi | Clinical-text lead, four-person research team
Led clinical-text modeling in a multimodal system predicting emergency-department diagnostic discordance from clinical text, physiological time series and chest X-rays across 50,967 aligned MIMIC-IV admissions.
Compared TF-IDF, BiLSTM and Clinical ModernBERT; contributed patient-level splitting, ablations, rationale extraction, explanation-faithfulness testing and calibration analysis.
The team's final model achieved AUROC 0.807 on 7,721 held-out admissions; the outcome was diagnostic discordance, not independently verified misdiagnosis.

Independent Medical Imaging Research, American University *Aug 2025 - Nov 2025*
Trained DenseNet-121 and EfficientNet-B0 for chest X-ray classification on 35,000 MIMIC-CXR images; evaluated zero-shot external transfer on 234 CheXpert images without target-domain fine-tuning.
DenseNet-121 achieved mean AUROC 0.764 internally and 0.797 on the small external sample; the limited external cohort constrains generalization claims.

TEACHING

Graduate Teaching Assistant, American University *Aug 2024 - Present*
Supported statistics, machine learning and data science instruction and assessment; developed Python tutorials on regression, probability, hypothesis testing and core machine-learning concepts.

RESEARCH OUTPUTS

Technical report (preliminary; not peer-reviewed)
Ozo-ogueji, P. C. (2026). “Continual Medical Image Classification on BMAD: Rehearsal, Regularization and Forgetting in a Domain-Incremental Stream.” Faculty mentor: Prof. Roberto Corizzo.

Manuscripts in Preparation
1. “Multimodal Deep Learning for Predicting Emergency Department Diagnostic Discordance Using MIMIC-IV Clinical Text, Time-Series, and Chest X-Ray Imaging Data” [code]
Four-person research team; led the clinical-text component and contributed calibration, interpretability and multimodal evaluation.

2. “Cross-Institutional Validation of Deep Learning for Multi-Disease Chest X-Ray Classification Using MIMIC-CXR and CheXpert” [code]
Independent medical-imaging study; manuscript in preparation.

SELECTED PROJECTS

Automated COVID-19 Misinformation Detection
Fine-tuned RoBERTa and Llama-3.2-1B-Instruct on CoAID; investigated class imbalance with weighting and oversampling and evaluated uncertainty with bootstrap resampling.

Healthcare Payment Variance Analytics and Contract Modeling
Built a reproducible Medicare analytics pipeline using Python, DuckDB, Pandera and Parquet to analyze hospital- and DRG-level payment variation.

Multimodal Hate Speech Detection
Combined BERT text and ResNet50 image representations; compared fusion strategies using F1-based evaluation.

PROFESSIONAL

Data Scientist, METY Technology *Feb 2024 - Present*
Developed and evaluated gradient-boosting models for health-related engagement outcomes using feature engineering, cross-validation and hyperparameter optimization; communicated findings to product leadership. Reviewed health-app research and analyzed feature combinations to inform product decisions.

Data Analyst, Polaris Bank Limited, Owerri, Nigeria *Feb 2018 - Aug 2023*
Supported fraud analytics and reporting workflows; developed ETL pipelines, data-warehouse components and executive dashboards.

TECHNICAL METHODS

Languages and frameworks: Python, R, SQL; PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, XGBoost, LightGBM, Mammoth
Methods: continual learning (replay, regularization, gradient constraints), domain-incremental evaluation, group-aware splitting, AUROC and forgetting analysis, multimodal fusion, external validation, calibration and bootstrap inference
Tools: Git/GitHub, Jupyter, Google Colab, AWS, DuckDB, Pandera, Spark, Databricks, PostgreSQL; PyCLAD (exploratory continual anomaly-detection work)

HONORS

$20,000 Merit Scholarship, American University
Google Hackathon Competition - Winning team member
Smith Analytics Consortium Fifth Annual Datathon - Participant
Fourth Penn Conference on Big Data in Biomedical and Population Health Sciences - Attendee
Graduate Behavioral Faculty Fellows Program, American University - Participant